

JAVA 8 MASTER HANDBOOK

Lecture 1 – Java 8 Introduction

Part 3 – Java 8 Features Overview

README.md (Section 3)

Chapter 11 - Nashorn JavaScript Engine

Java 8 me ek aur feature introduce hua tha.

Iska naam tha

Nashorn JavaScript Engine

Question.

Java ke andar JavaScript kyun?

Us time bahut saari enterprise applications me

Java aur JavaScript

ek saath use hote the.

Developers chahte the

ki Java application ke andar hi

JavaScript code execute kiya ja sake.

Isi requirement ko dhyan me rakhkar

Java 8 me

Nashorn JavaScript Engine

introduce kiya gaya.

Purpose

- Java ke andar JavaScript execute karna.
 - Scripting support provide karna.
 - Dynamic execution ko support karna.
-

Example Use Cases

- Configuration Scripts
 - Automation
 - Embedded JavaScript Logic
 - Rule Engines
-

Important Interview Point

Question

Is Nashorn important today?

Answer

Generally **No**.

Reason

- Java 8 me introduce hua.
- Later Java versions me deprecated hua.
- Baad me remove bhi kar diya gaya.

Isliye modern projects me rarely use hota hai.

Interview me sirf itna batana kaafi hota hai.

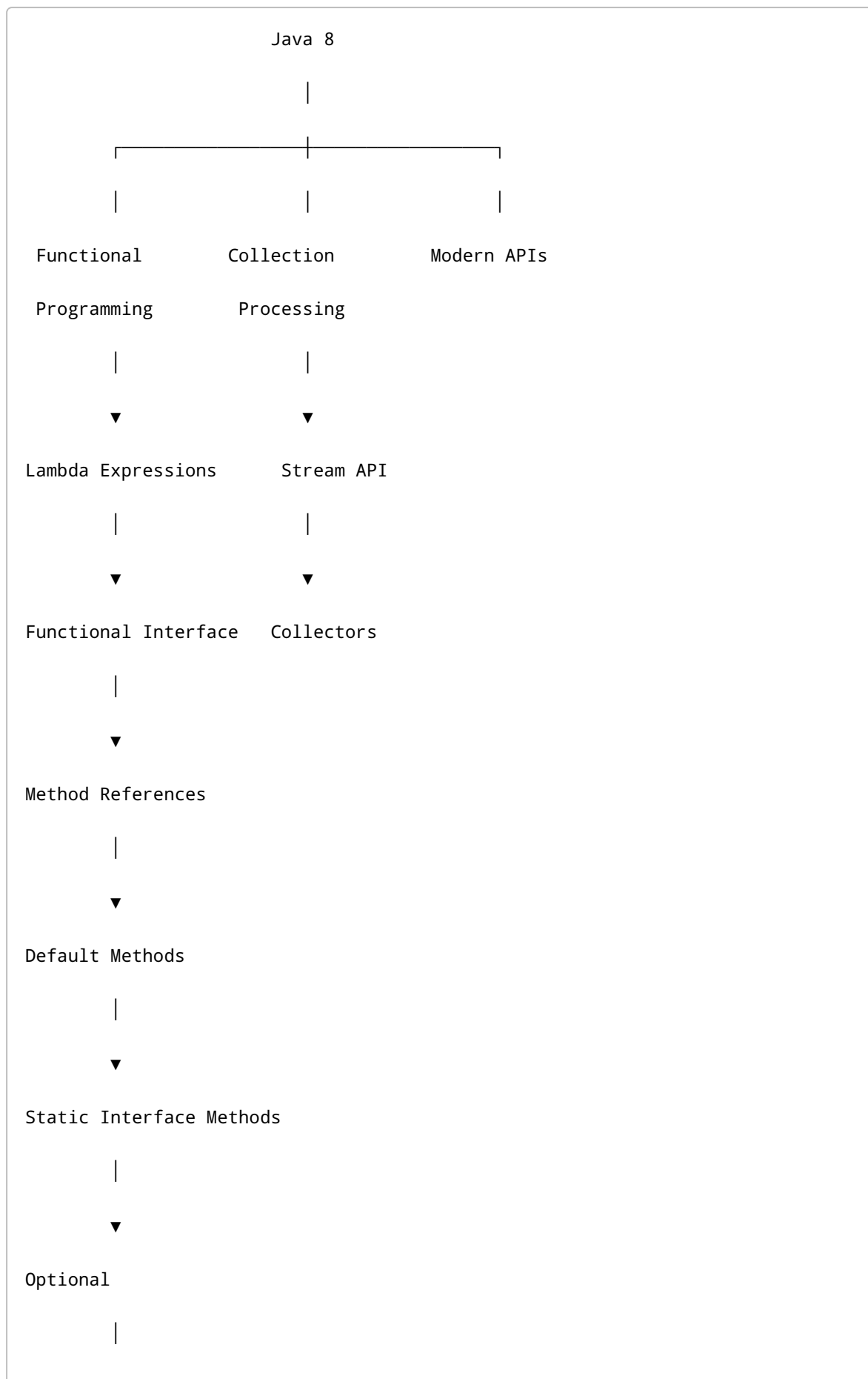
Chapter 12 - Java 8 Feature Dependency Flow

Ab tak humne jitne features padhe

wo actually alag-alag features nahi hain.

Ye ek complete ecosystem hai.

Diagram dekho.





Date & Time API

Observe carefully.

Poor Java 8

random features ka collection nahi hai.

Ye ek connected ecosystem hai.

Relationship Between Features

Sabse important relationship.

Lambda Expression



Implements



Functional Interface



Used By



Stream API



Produces



Result

Ye diagram

Java 8 ka backbone hai.

Agar ye samajh aa gaya

to poora Java 8

bahut aasaan ho jayega.

Java 8 Learning Roadmap

Ye wahi roadmap hai

jo hum is course me follow karenge.

Java 8 Introduction

↓

Lambda Expressions

↓

Functional Interfaces

↓

Method References

↓

Default Methods

↓

Predicate

↓

Function

↓

Consumer

↓

Supplier

↓

Stream API

↓

Collectors

↓

Optional

↓

Date & Time API

↓

Parallel Streams

↓

Interview Programs

Is sequence ko intentionally design kiya gaya hai.

Agar sequence tod diya

to concepts difficult lagne lagenge.

Chapter 13 - Which Features Are Most Important?

Interview Priority

Feature	Priority	Interview Frequency
Lambda Expressions	★★★★★★	Very High
Functional Interfaces	★★★★★★	Very High
Stream API	★★★★★★	Very High
Optional	★★★★★★	High
Date & Time API	★★★★★★	High
Method References	★★★★★	High
Default Methods	★★★★★	Medium
Static Interface Methods	★★★★	Medium
Nashorn	★★★	Low

Which Feature Should Be Learned First?

Bahut students directly

Stream API

start kar dete hain.

Ye galti hai.

Correct Order

Lambda

↓

Functional Interface

↓

Method Reference

↓

Predicate

↓

Function

↓

Consumer

↓

Supplier

↓

Stream API

Isliye hum bhi

isi order me padhenge.

Common Misconceptions

Misconception 1

✗ Stream data store karti hai.

✓ Correct

Stream

sirf data ko process karti hai.

Data

Collection me hi rehta hai.

Misconception 2

✗ Lambda ek function hai.

✓ Correct

Java me Lambda

Function-like syntax hai

jo Functional Interface ke abstract method ki implementation represent karta hai.

Misconception 3

✗ Optional use karne se NullPointerException kabhi nahi aata.

✓ Correct

Optional

null ko represent karne ka better mechanism hai.

Ye har situation me

NullPointerException ko eliminate nahi karta.

Misconception 4

✗ Default Methods Multiple Inheritance provide karti hain.

✓ Correct

Default Methods

Interface evolution aur backward compatibility ke liye introduce ki gayi thi.

Multiple inheritance ke kuch scenarios ko support karti hain,

lekin Java ko C++ jaisa multiple inheritance language nahi bana deti.

Misconception 5

✗ Streams hamesha faster hoti hain.

✓ Correct

Har situation me nahi.

Performance

problem aur data size

par depend karti hai.

Hum ye Parallel Streams lecture me detail me padhenge.

Interview Questions

Q1. Name all major features introduced in Java 8.

Answer

- Lambda Expressions
- Functional Interfaces
- Stream API
- Method References
- Default Methods
- Static Interface Methods
- Optional
- Date & Time API
- Nashorn JavaScript Engine

Q2. Which Java 8 feature is the foundation of Stream API?

Answer

Lambda Expressions

aur

Functional Interfaces.

Q3. Why was Stream API introduced?

Answer

Collections ko

Declarative Programming style me

process karne ke liye.

Q4. Why were Default Methods introduced?

Answer

Existing interfaces ko

without breaking

evolve karne ke liye.

Q5. Why was Optional introduced?

Answer

Null handling improve karne ke liye.

Absent values ko

explicitly represent karne ke liye.

Q6. Why was the New Date-Time API introduced?

Answer

Old Date API

mutable thi.

Thread-safe nahi thi.

Confusing thi.

Java 8 ne

immutable

thread-safe

aur modern API provide ki.

Q7. Which Java 8 feature is most important for interviews?

Answer

- Lambda Expressions
- Functional Interfaces
- Stream API

Ye teen sabse important hain.

Q8. Can Streams store data?

Answer

No.

Streams

sirf process karti hain.

Store nahi karti.

Q9. Is Lambda an Object?

Answer

Lambda expression khud object nahi hota.

Lekin runtime par JVM usse Functional Interface ke implementation ke roop me represent karti hai.

Is topic ko hum Lambda Internals me detail me padhenge.

Q10. Is Java a Functional Language?

Answer

No.

Java

primarily

Object-Oriented Programming language hai.

Java 8 ke baad

Functional Programming features support karta hai.

Part 3 Summary

Aaj humne Java 8 ke major features ka overview dekha.

Important points:

- Java 8 ek ecosystem hai.
 - Lambda Expressions foundation hain.
 - Functional Interfaces Lambda ko support karti hain.
 - Method References cleaner syntax provide karti hain.
 - Stream API collection processing ke liye hai.
 - Default Methods backward compatibility ke liye hain.
 - Static Interface Methods utility methods ke liye hain.
 - Optional null handling improve karta hai.
 - Date-Time API old Date classes ka replacement hai.
 - Nashorn JavaScript Engine JavaScript execution ke liye introduce hua tha.
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Quick Revision Sheet

Java 8

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Lambda

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Functional Interface

↓

Method Reference

↓

Predicate

↓

Function

↓

Consumer

↓

Supplier

↓

Stream API

↓

Collectors

↓

Optional

↓

Date-Time API

↓

Is flow ko yaad rakhna.

Ye poore Java 8 course ka roadmap hai.

Homework

1. Java 8 Feature Dependency Diagram khud banao.
 2. Java 8 ke saare features likho aur unka purpose ek line me explain karo.
 3. Lambda aur Functional Interface ka relation explain karo.
 4. Stream API data process karti hai ya data store? Explain.
 5. Optional kis problem ko solve karta hai?
 6. Default Methods ka primary objective kya tha?
 7. Nashorn JavaScript Engine aaj production me kam kyun use hota hai?
 8. Java 8 ke top 5 interview features likho.
 9. "Java 8 is an ecosystem, not just a collection of features." Explain.
 10. Java 8 learning roadmap ko bina notes dekhe likhne ki practice karo.
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End of Lecture 1 - Part 3

Agar tum Part 1, Part 2 aur Part 3 ko achchhe se samajh lete ho,

to tumhare paas Java 8 ki foundation ready hai.

Ab hum **Lecture 1 – Part 4** me poore Lecture 1 ka conclusion karenge, jisme:

- Java 7 vs Java 8 (Complete Comparison)
- Real Industry Use Cases
- Top Interview Questions
- Java 8 Architecture
- Common Mistakes
- Final Cheat Sheet
- Master Revision Notes
- Lecture 1 Final Summary